

PRIYANKA YADAV

Machine Learning / AI Engineer · MS Computer Science

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Open to Relocate | H4 - EAD

PROFESSIONAL SUMMARY

M.S. Computer Science, University of Illinois Springfield (GPA 3.87/4.00), specializing in machine learning, deep learning, computer vision, NLP, and transformer architectures. Hands-on experience building end-to-end ML systems — data preprocessing, model training, evaluation, and deployment — using PyTorch, Hugging Face, LangChain, and FastAPI, across 3 independent projects spanning computer vision, NLP, and LLM/RAG applications. Prior experience as a data analyst translating stakeholder requirements into technical specifications alongside engineering teams, with strong cross-functional communication skills applicable to client-facing and consulting environments. Seeking Machine Learning Engineer, AI Engineer, Data Scientist, or AI/ML Consultant roles.

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript, Java
ML / Deep Learning:	PyTorch, TensorFlow (basic), scikit-learn, CNNs, Transfer Learning (ResNet, VGG), Object Detection (YOLO, Faster R-CNN), Image Segmentation (U-Net, Mask R-CNN), Dropout, BatchNorm
Computer Vision:	OpenCV, Image Classification, Object Detection, Image Segmentation, Data Augmentation, Model Evaluation (AUC, Sensitivity, Specificity)
NLP & LLMs:	Tokenization, TF-IDF, Word2Vec, RNN/LSTM/GRU, Transformers (Self-Attention, Multi-Head Attention, Positional Encoding, Encoder-Decoder), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI Systems, GPT architecture
Libraries:	NumPy, Pandas, Matplotlib, Seaborn, Hugging Face, SciPy, LangChain, ChromaDB
Cloud & Tools:	Git, GitHub, Docker, AWS, Streamlit, FastAPI, Jupyter
Data:	Data preprocessing, feature engineering, handling imbalanced datasets, PostgreSQL, MongoDB

PROJECTS

Independent, end-to-end ML builds applying computer vision, NLP, and LLM/RAG techniques across different problem domains — from data pipeline through deployment.

Dental Disease Detection from Panoramic X-Rays

Computer Vision · YOLOv8 · MobileNetV2 · Transfer Learning · PyTorch · Streamlit · FastAPI · Docker

- Built a two-model computer vision pipeline for automated dental disease detection on panoramic (OPG) X-rays: a MobileNetV2 classifier for cropped tooth images and a YOLOv8 detector for full-image tooth localization (FDI numbering) and diagnosis classification (caries, deep caries, periapical lesion, impacted tooth).
- Trained and evaluated on the DENTEX dataset (MICCAI 2023 challenge) with image preprocessing, augmentation, and train/val/test splits; detector currently reaches mAP50 of 0.48, with ongoing work to improve rarer classes affected by class imbalance.
- Deployed as a live demo backed by a FastAPI service, containerized with Docker.
- Live demo: caries-detection-rdtqwvaq9lnuov6gspyyqc.streamlit.app · Code: github.com/PriyankaYadav23/Caries-Detection

Medical Coding & Clinical Decision Support Assistant — NLP + Agentic RAG (ICD-10 / CDT)

Python · PyTorch · Hugging Face (BioBERT, SapBERT, BGE) · Groq LLM · FastAPI · Streamlit · Docker

- Built a weak-labeling pipeline with SapBERT embeddings to map 4,999 unlabeled clinical notes to the 74,719-code ICD-10-CM codebook, retaining 99.8% of notes vs. 75% with a TF-IDF baseline
- Fine-tuned BioBERT on 232 ICD-10 codes (3,569 notes); used per-class metrics to find a rare-code recall gap and applied class-weighted loss, raising macro recall from 0.88 to 0.92 and macro F1 from 0.86 to 0.89.
- Built a RAG retriever over the official CDC ICD-10-CM guidelines (121 pages → 365 chunks, BGE embeddings); validated it with known-answer queries and fixed a ranking error through an A/B-tested query instruction.
- Designed a tool-calling LLM agent (classifier, guideline search, code lookup) with guardrails: prompt constraints, step limits, retries and Python verification of every output code, which caught the fabricated codes, confidence scores and citations seen in the unguarded version.

- Served the classifier and agent through FastAPI (/predict, /analyze) with a Streamlit review UI; containerized the classifier API with Docker.
- Code: github.com/PriyankaYadav23/ICD-10-Coding-Agent

Adaptive Practice Recommender — Knowledge Tracing

Classic ML · Random Forest · XGBoost · Feature Engineering · scikit-learn

- Built a predictive model on a 100M+ row real-world interaction-log dataset (EdNet) to estimate a student's probability of answering correctly from historical behavior — a feature-engineering and classic-ML workflow generalizable to churn, risk, or recommendation problems.
- Engineered features from raw interaction logs; benchmarked Logistic Regression, Random Forest, and XGBoost.
- Live demo: adaptive-practice-recommender-2vlca9ytfpgsww5ubefrm9.streamlit.app · Code: github.com/PriyankaYadav23/Adaptive-Practice-Recommender

PROFESSIONAL EXPERIENCE

Clinical Data Analyst — CBCT & Panoramic Imaging

Aug 2021 – Nov 2022

Dr. Hemant CBCT Center · Udaipur, Rajasthan, India

- Handled clinical imaging data: acquired CBCT and panoramic scans, and managed DICOM files and their metadata in diagnostic imaging software, using DICOM viewers and PACS.
- Reviewed imaging studies for quality and diagnostic accuracy to support treatment planning for referring dentists, using DICOM imaging software.
- Kept patient scan records accurate and complete (patient details, scan type, referring provider), and pulled records with queries, using SQL and MySQL.
- Tracked referral sources and scan volumes to spot trends and guide outreach to local dentists and specialists, using SQL and Power BI.

Data Analyst Intern — Health Gennie

Mar 2021 – Jul 2021

Fitkid Health Tech Pvt Ltd · Jaipur, Rajasthan, India

- Researched and defined the clinical decision logic for a remote patient-monitoring platform (vitals thresholds and symptom-based routing to the right specialty), using Excel and Python (Pandas).
- Turned clinical needs into product requirements and information architecture (data fields, patient workflows, referral rules) with the engineering team, using SQL and MySQL (database design).
- Wrote documentation that connected clinical terms to how the platform handled patient vitals data, using Google Docs.
- Onboarded physicians and gathered their feedback on workflows, which helped improve the platform's usability for providers, using Excel and Google Sheets.
- Helped prepare investor pitch materials summarizing the platform's clinical value and digital health use case, using PowerPoint and Excel charts.

EDUCATION

M.S., Computer Science

Aug 2024 – Jul 2026

University of Illinois Springfield · USA

- GPA: 3.87/4.00. Coursework: Machine Learning, Advanced Database Concepts, Big Data Analytics, Database Systems, Web Design, Java GUI Development, Graduate Research Seminar, Graduate Programming Practicum.
- Independent study: Deep Learning, Computer Vision, NLP, Transformers.

MBA, Healthcare Management

2019 – 2021

IIHMR University · India

- Focus: health informatics, hospital administration, digital health strategy, health economics.

Bachelor of Dental Surgery (BDS)

2013 – 2018

Pacific Dental College & Hospital · India

- 5-year integrated clinical program, including compulsory rotatory clinical internship at the college hospital — oral radiology, panoramic X-ray interpretation, endodontics, oral surgery, prosthodontics, pediatrics.